

A Finite-Sample Obstruction to Quotient Uniformity

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Abstract

RH-259 finds contractive twelfth powers at 23 finite quotient endpoints. We prove that such finite pointwise data, without a parameter modulus or interval family enclosure, cannot imply a uniform small-noise theorem. Nine archived endpoints also remain uncomputed. This is a logical insufficiency result for the present evidence, not a proof that the underlying quotient family is nonuniform.

1 The finite-sample theorem

Theorem 1.1 (No uniform supremum from finite samples). *Let F be a finite subset of an interval I . For any prescribed values $v_x < 1$ at $x \in F$, there is a continuous function $f : I \rightarrow \mathbb{R}$ with $f(x) = v_x$ on F and $\sup_I f > 1$.*

Proof. Interpolate the finite prescribed values by a continuous function g . Choose $x_0 \in I \setminus F$ and a triangular bump b supported in a neighborhood of x_0 disjoint from F . Then $g + Cb$ retains every sampled value, while a sufficiently large C makes its value at x_0 exceed one. $\square$

The theorem remains true if all sampled values lie below a common $q < 1$. A uniform conclusion becomes possible only after additional structure controls the values between samples, for example a certified Lipschitz modulus or a single interval-operator enclosure.

2 Application to the quotient ledger

The archived RH-259 data [2] contain 23 endpoints at 13 distinct noise values: 10 left and 13 right. All 23 twelfth powers are contractive, while no one-step block is contractive. The first contractive depth ranges from 3 to 9, and

$$0.22185212659640824 \leq q_{12} \leq 0.5056418005507071.$$

The finite unit-disk tail diagnostic is $5.654507945432548 \times 10^{-4}$. These are finite matrix findings, not interval enclosures.

Corollary 2.1 (Scoped route stop). *The 23 RH-259 contractions alone do not establish a uniform quotient-tail bound on the small-noise continuum.*

Proof. Apply the theorem to the unsampled parameter values. Independently, 9 of the 32 archived endpoints are not present in RH-259, so even finite archived coverage is incomplete. $\square$

This does not contradict the conditional block-power theorem of RH-246 [1]; it identifies missing hypotheses needed to apply it uniformly. It does not prove nonuniformity of the actual family. The cloud bridge and Gates A–E remain false/open; no Hilbert–Polya, zeta-divisor, or RH claim is made.

References

- [1] Bin Wang. Block-power quotient envelope criterion. *Preprint*, 2026. RH-246 preprint.
- [2] Bin Wang. Extended quotient block-power diagnostic. *Preprint*, 2026. RH-259 preprint.